FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Predictive Maintenance and Anomaly Detection Algorithm for Biomethane Plants

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Mestrado em Engenharia e Ciência de Dados

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July 12, 2026

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Declaration of Honor

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Abstract

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Acknowledgements

Write your acknowledgements here. Do not forget to mention projects and/or grants that you have benefited from, institutions or companies that may have provided for conditions to enable or make your work more amenable, if any. Ask your supervisor about the specific textual format to use, as funding agencies are often strict about this.

Author

"Until I began to learn to draw,
I was never much interested in looking at art."

Richard P. Feynman

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Contents

List of Figures

List of Tables

Abbreviations

AI	Artificial Intelligence
DL	Deep Learning
ML	Machine Learning
EDA	Exploratory Data Analysis
CV	Computer Vision
NLP	Natural Language Processing
PdM	Predictive Maintenance
AD	Anomaly Detection
FD	Fault Detection
CPS	Cyber-Physical Systems
GPU	Graphics Processing Unit
TPU	Tensor Processing Unit
IoT	Internet of Things
PSA	Pressure Swing Adsorption
VPSA	Vaccum Pressure Swing Adsorption
TSA	Temperature Swing Adsorption
ESA	Electric Swing Adsorption

Chapter 1

Introduction

1.1 Context

The 21st century has witnessed remarkable advances in Artificial Intelligence (AI) systems, reshaping numerous fields, including the industrial and defense sectors (**?**). The introduction of AlexNet in 2012 (**?**), AlphaGo's victory over Lee Sedol in 2016 (**?**), and the public release of ChatGPT in 2022 (**?**) are among the most significant milestones in modern AI, advancing the fields of Computer Vision (CV), Sequential Decision-Making, and Natural Language Processing (NLP), respectively. These breakthroughs were driven by a rapid growth in computational power, the development of specialized hardware such as GPUs and TPUs, and the widespread adoption of Deep Learning (DL) techniques (**?**).

As a result, DL has become the dominant paradigm in modern Machine Learning (ML) due to its ability to learn highly complex and abstract representations directly from data (**?**).

1.2 The role of AI in the industry sector

Over the last fifteen years, the industrial sector has undergone a profound digital transformation, leading to the emergence of the Industry 4.0 paradigm. Industry 4.0 is characterized by the integration of advanced manufacturing technologies, Cyber-Physical Systems (CPS), the Internet of Things (IoT), cloud computing, and interconnected production environments. These technologies enable the continuous acquisition of large volumes of sensor data, creating smart factories capable of monitoring, communicating, and adapting their operation in real time.

The availability of these large-scale industrial datasets has created favourable conditions for the widespread adoption of DL techniques in this field. AI systems are increasingly being employed to support production planning and scheduling, process monitoring and control, Predictive Maintenance (PdM), robotics, quality inspection, and process optimization, contributing to improved efficiency, reliability, and productivity in modern manufacturing environments (**?**).

1.3 Motivation

Despite the significant advances in AI technologies, the application of DL techniques in Anomaly Detection (AD), Fault Detection (FD), and PdM tasks for biomethane plants remains limited. This limitation creates a gap in the literature regarding the development of advanced data-driven solutions for this industrial domain. Furthermore, there is a lack of research on multitask learning models capable of performing different types of predictions simultaneously in this industry. Therefore, the development and evaluation of such models represent a promising research direction. This thesis aims to train and evaluate different DL models (both multi-task and single-task) in order to investigate whether deep multitask learning approaches can improve prediction performance regarding AD, FD, and PdM when compared with traditional ML models.

1.4 Objectives

The objectives of this thesis are:

- To perform an Exploratory Data Analysis (EDA) of the biomethane upgrading dataset in order to better understand the data, characterize the process variables and identify potential challenges for model development.
- To develop and train different single-task DL models for anomaly detection, fault detection, and predictive maintenance tasks.
- To develop and train different multi-task DL models using multiple architectural configurations capable of jointly performing several prediction tasks.
- To develop and train traditional ML models for each prediction task, providing a baseline for comparison.
- To compare and evaluate the performance of all proposed models using appropriate evaluation metrics.
- To evaluate the robustness of the developed models under sensor noise.
- To investigate whether multi-task learning improves predictive performance compared with single-task DL models and traditional ML approaches.

1.5 Company Presentation

1.6 Theasis Structure

Chapter 2

Background

2.1 What is Biogas?

Biogas is a renewable fuel produced through the anaerobic digestion of biodegradable organic matter, a biological process in which microorganisms decompose organic compounds in the absence of oxygen. This process converts complex organic materials into simpler compounds while producing a gaseous mixture known as biogas. A wide range of feedstocks can be used for anaerobic digestion, including agricultural residues, livestock manure, municipal solid waste, sewage sludge, food waste, and other types of organic biomass, as illustrated in Figure **??**.

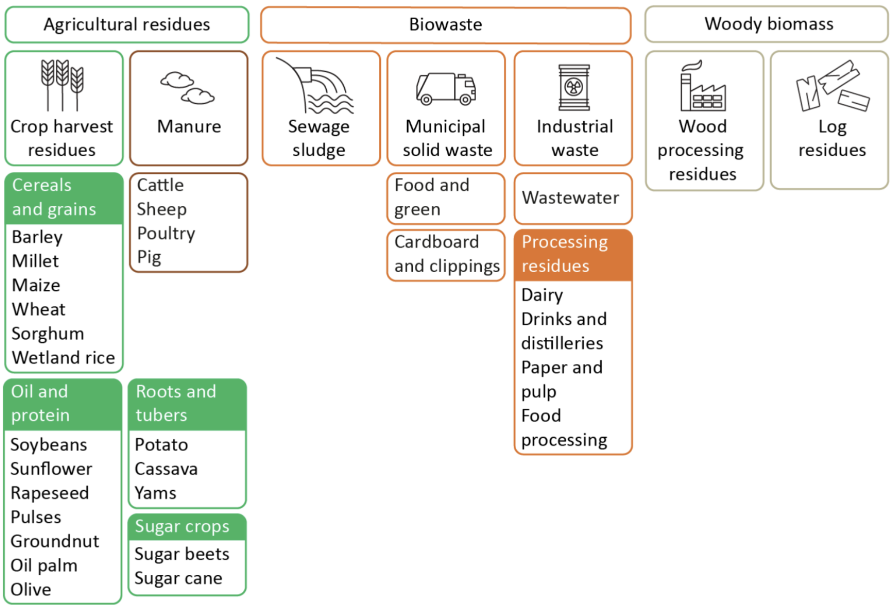


Figure 2.1: Sources of organic waste feedstock for biogas production. Source: **?**

The composition of biogas depends on the type of feedstock and the operating conditions of the anaerobic digestion process. It is primarily composed of methane (CH_4), typically accounting for 50–80% of its volume, and carbon dioxide (CO_2), which represents approximately 20–50%. In addition to these major constituents, biogas contains smaller concentrations of impurities, including hydrogen sulfide (H_2S), water vapour (H_2O), ammonia (NH_3), oxygen (O_2), and siloxanes.

Since methane is the component responsible for the calorific value, the relatively high concentration of carbon dioxide and other contaminants limits the direct use of raw biogas in many applications. Consequently, purification processes are required to increase its methane content and produce biomethane with properties comparable to those of natural gas (**?**).

2.2 Biogas Upgrading

Biogas upgrading, also referred to as biogas purification, biogas refinement or biogas methanation, is the process of increasing the methane concentration of raw biogas by removing carbon dioxide and other undesirable contaminants. The resulting methane-rich gas, known as biomethane, has a significantly higher calorific value and can be used as a substitute for natural gas in a wide range of applications. In general, biomethane should contain at least 95% methane to comply with the quality requirements for grid injection and other energy applications (**?**).

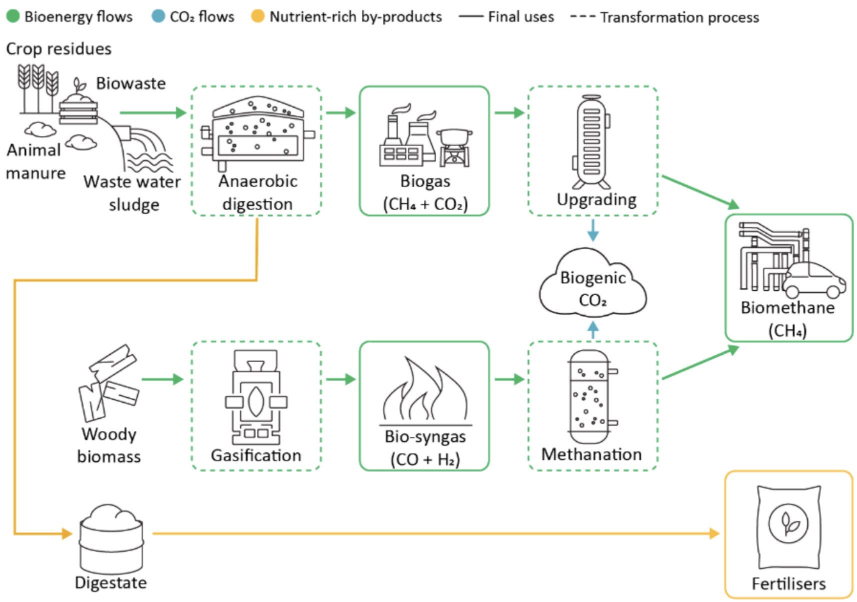


Figure 2.2: Biogas upgrading and utilization. Source: **?**

Several upgrading technologies have been developed to achieve this objective, each presenting different advantages, limitations, and operating requirements. Since a detailed comparison of these technologies is beyond the scope of this dissertation, only a brief overview of the main biogas upgrading methods is provided in the following section.

2.2.1 Biogas Upgrading Technologies

The most widely adopted methods can be broadly classified into adsorption, membrane separation, cryogenic separation, and absorption processes (**?**):

- **Adsorption separation:** One of the most widely used methods for biogas upgrading. Adsorption processes can be further classified into Pressure Swing Adsorption (PSA), Temperature Swing Adsorption (TSA), and Electric Swing Adsorption (ESA), with PSA being the most widely adopted. In PSA systems, biogas is compressed and passed through one or more columns filled with an adsorbent material. The adsorbent selectively retains contaminants while methane passes through the column. Once saturated, the column is regenerated by depressurization, allowing the adsorbed gases to be released before starting a new adsorption cycle.

- **Membrane separation:** This method uses semi-permeable membranes, typically made of polymeric materials, to separate methane from carbon dioxide and other impurities according to their different permeation rates. The membrane material and configuration depend on the desired biomethane purity.

- **Cryogenic separation:** Cryogenic upgrading exploits the different boiling points of the gas components. By progressively reducing the temperature, carbon dioxide and other contaminants condense before methane, allowing their separation.

- **Absorption separation:** Absorption processes use a liquid solvent, such as water or chemical solvents, to selectively absorb carbon dioxide from biogas. Water scrubbing is one of the most common upgrading technologies, taking advantage of the significantly higher solubility of carbon dioxide compared to methane under standard operating conditions.

Figure **??** summarizes the four main biogas upgrading technologies discussed in this section.

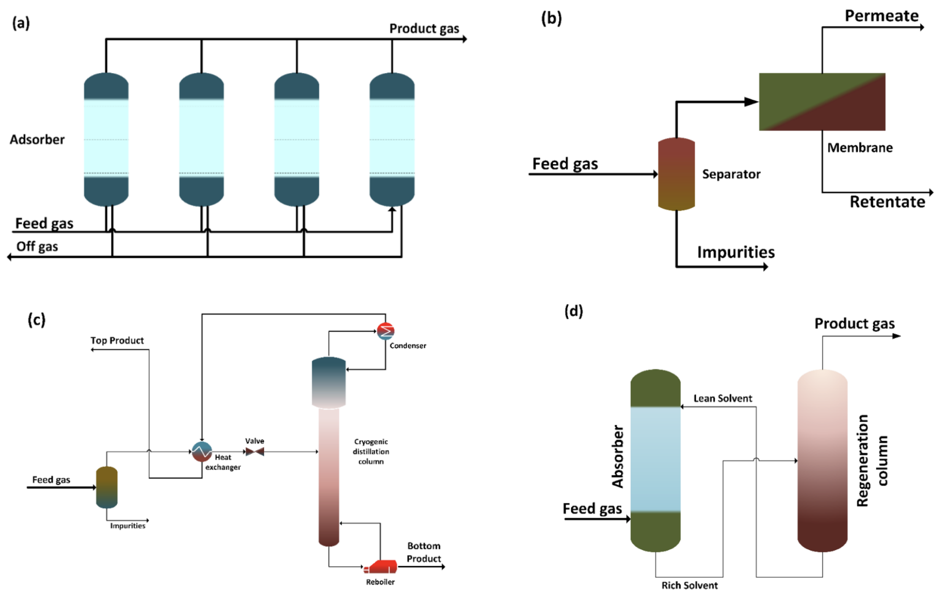


Figure 2.3: Main biogas upgrading technologies: adsorption, membrane separation, cryogenic separation, and absorption. Source: **?**.

Among the available technologies, a PSA biomethane upgrading plant was considered in this dissertation. Since the proposed AI models are developed using data collected from a PSA-based upgrading system, the following section presents a more detailed description of its operating principles and process configuration.

2.3 Considered Biomethane Plant

As discussed in the previous section, the biomethane upgrading plant considered in this dissertation employs PSA technology. More specifically, the plant operates using a four-column adsorption cycle and incorporates a vacuum pump during the regeneration stage to enhance the desorption of the adsorbed gases. This configuration is commonly referred to as Vacuum Pressure Swing Adsorption (VPSA).

A photograph of an example biomethane upgrading plant used in this work is presented in Figure **??**.

Figure 2.4: Industrial VPSA biomethane upgrading plant (adapted from Sysadvance)

2.3.1 VPSA Process

VPSA separates gas components by exploiting their different affinities for porous adsorbent materials under varying pressure conditions. As previously explained, raw biogas is primarily composed of methane (CH_4) and carbon dioxide (CO_2), together with smaller amounts of water vapour and other contaminants. Before entering the adsorption unit, the biogas must first be dried to remove the water vapour. This step is essential because the adsorbent materials used inside the adsorption columns, one of which is ceramic-based, exhibit a strong affinity for water. Since the adsorption process is exothermic, excessive moisture can lead to overheating and permanent damage to the adsorbent material.

After the drying stage, the biogas is compressed before entering the adsorption columns. Operating at elevated pressures increases the probability of gas molecules interacting with the adsorbent material, improving the separation efficiency. Each adsorption column is filled with adsorbent materials specifically selected to preferentially retain carbon dioxide while allowing methane to pass through. As a result, a methane-rich stream leaves the column and is collected as biomethane.

As the adsorption process continues, the adsorbent gradually becomes saturated with carbon dioxide and loses its separation capacity. At this point, the column must be regenerated before it can be reused. Instead of stopping the production process, VPSA systems employ multiple adsorption columns operating simultaneously at different stages of the cycle. In the plant considered in this work, four adsorption columns operate in a synchronized sequence, ensuring continuous biomethane production while one or more columns undergo regeneration.

Figure **??** illustrates the operating principle of the four-column VPSA process. During the adsorption stage, compressed biogas enters one of the columns, where carbon dioxide is selectively adsorbed while methane exits as the product gas. Once the adsorbent approaches saturation, the column undergoes a pressure equalization stage, in which part of the compressed gas is transferred to another column operating at a lower pressure. This step recovers part of the compression energy and improves the overall efficiency of the process.

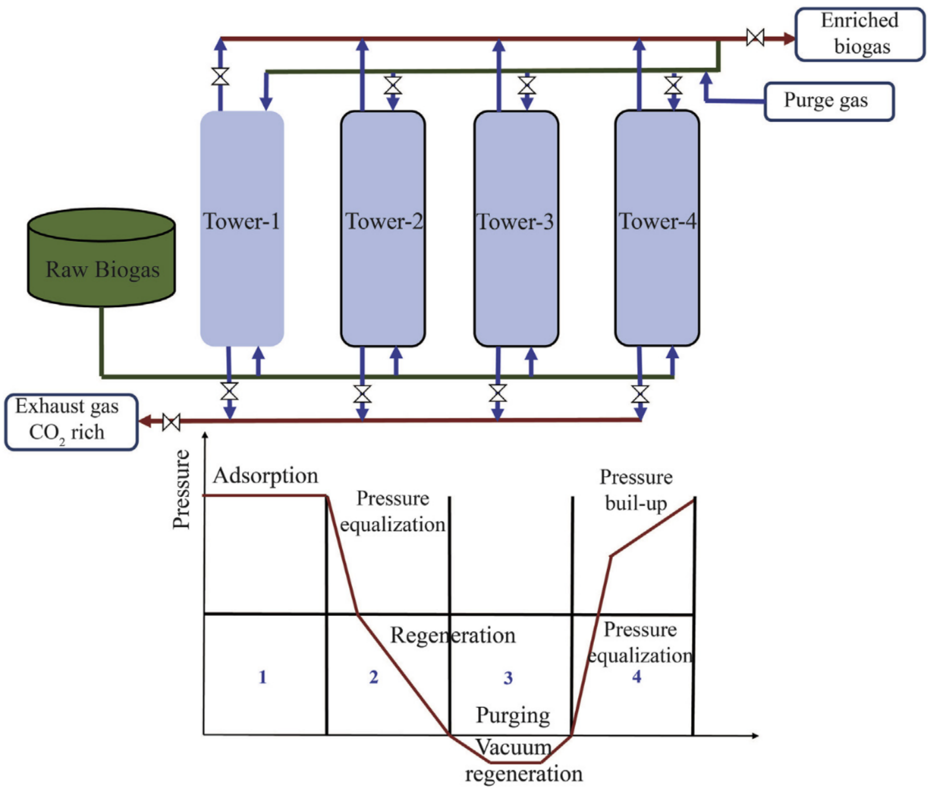


Figure 2.5: Operating cycle of a four-column VPSA system. Source: **?**

The column is then depressurized and enters the regeneration stage. Unlike conventional PSA systems, VPSA technology employs a vacuum pump to further reduce the pressure below atmospheric conditions. The low pressure promotes the desorption of the retained carbon dioxide, regenerating the adsorbent material and restoring its adsorption capacity. Since a small amount of methane is also desorbed during the initial seconds of the vacuum stage, this gas is recovered and redirected to the biogas inlet to minimize methane losses.

Finally, the regenerated column is gradually repressurized through another pressure equalization step, followed by pressure build-up until the operating pressure is reached. The column is then ready to begin a new adsorption cycle. By continuously alternating the operating stage of each column, the VPSA system is able to produce biomethane without interrupting the upgrading process.

In practice, only one adsorption column is in the production stage at any given instant, while the remaining columns are simultaneously undergoing pressure equalization, regeneration, or repressurization (**?**).

2.3.2 Vacuum Pump

As discussed in the previous section, the vacuum pump plays a fundamental role in the VPSA process by enabling the regeneration of the adsorption columns. By reducing the column pressure below atmospheric conditions, the vacuum pump promotes the desorption of the retained gases, restoring the adsorption capacity of the adsorbent material and allowing the adsorption cycle to continue. Consequently, the performance of the vacuum pump has a direct impact on the efficiency, productivity, and methane recovery of the entire biomethane upgrading process.

Due to its critical role in the plant operation, the vacuum pump is extensively monitored in the considered biomethane plant. Several process variables, including pressure, temperature, electrical current, power consumption, and rotational speed, are continuously acquired to assess its operating condition and detect potential abnormal behaviour.

Since this dissertation focuses on the development of AI-based models for anomaly detection, fault detection, and predictive maintenance, the vacuum pump was selected as the primary component under analysis. Figure **??** shows a photograph of a working vacuum pumo inside a biogas upgrading plant.

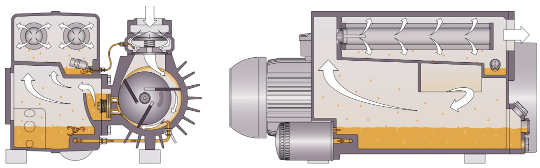
Figure 2.6: Vacuum pump inside a biogas upgrading plant. Adapted from Sysadvance

2.3.2.1 Components

The vacuum pump used in the considered biogas upgrading plant, whose manufacturer cannot be disclosed due to confidentiality agreements, operates according to the rotary vane principle. Oil plays a key role in the pump operation by sealing the clearances between moving parts, providing lubrication, and dissipating the heat generated during gas compression. Figure **??** illustrates the operating principle, providing the necessary background to understand the vacuum process. Gas enters from the top, passes through the vanes.

A schematic representation of the pump is also provided in Figure **??**, where the main internal components are identified. Since a detailed mechanical description of the pump is beyond the scope of this dissertation, only the components that are relevant for understanding the monitored variables and the analysed failure modes are discussed.

[Vacuum pump operating principle.]



[Vacuum pump schematic.]

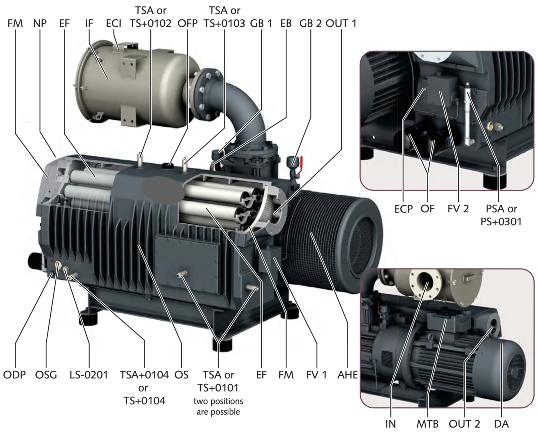


Figure 2.7: Vacuum pump used in the considered VPSA system.

Chapter 3

Another Chapter

3.1 What is an AI System?

According to the AI Act Regulation (EU) 2024/1689 of the European Parliament [insert ref], we can define an AI system as a system that is machine-based, can operate with some degree of autonomy, may show signs of adaptiveness after its deployment, and from a given input it recieves, generates an output that can influence its enviroment. These outputs can be predictions, content, recommendations, or decisions.

From a academic perspective, the authors from [insert ref] define AI in terms of intelligent agents, describing them as systems that perceive their environment through sensors and act upon that environment through actuators. Under this view, an AI system can be regarded as a function that maps observations of the environment to actions intended to achieve specific goals.

Although these definitions originate from different contexts, they share the same fundamental idea that an AI system recieves some kind of input, process it, and outputs something in order to achieve a specific goal.

Chapter 4

Conclusions and Future Work

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4.1 Results

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4.2 Further Work

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4.3 Impact on Sustainable Development Goals

Add in this section a reflection on how your work impacts the United Nations's Sustainable Development Goals (SDG). If you want to read about the SDG, visit the page: `https://sdgs.un.org/goals`. To obtain more information on the subject, we suggest the following courses:

- `ttps://dl4sd.org/course/view.php?id=4`
- `https://www.classcentral.com/course/global-sustainable-development-10311`

Appendix A

Loren Ipsum

If you are going to use a passage of Lorem Ipsum, you need to be sure there isn't anything embarrassing hidden in the middle of text.

A.1 What is *Loren Ipsum*?

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum[1].

A.2 Where does Loren come from?

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. T he first line of Lorem Ipsum, "Lorem ipsum dolor sit amet…", comes from a line in section 1.10.32.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

[1] Available at `http://www.lipsum.com/`

A.3 Why using Loren?

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using “Content here, content here”, making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for “lorem ipsum” will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

A.4 Where to Find Examples?

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by injected humour, or randomised words which don’t look even slightly believable. If you are going to use a passage of Lorem Ipsum, you need to be sure there isn’t anything embarrassing hidden in the middle of text. All the Lorem Ipsum generators on the Internet tend to repeat predefined chunks as necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with a handful of model sentence structures, to generate Lorem Ipsum which looks reasonable. The generated Lorem Ipsum is therefore always free from repetition, injected humour, or non-characteristic words etc.

Appendix B

Transparency on the Use of AI

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